

Does Expert Sparsity Concentrate Social Bias?

A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant* at 0.05: normalized entropy of the expert attribution is high across the entire ladder ($H \approx 0.79\text{--}0.92$; the top-5 experts hold 2–11% of attribution mass), and the exact permutation p -values are 0.106 (all 6 rungs, $\rho_H = +0.754$), 0.100 (5 rungs), 0.167 (4 rungs), and 0.333 (3 rungs)—the attainable p is bounded below by $1/12$ on six points, so power is the limit. Dense baselines sit clearly lower ($H \in [0.63, 0.76]$, mean 0.69), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Remaining flags are limited to the null-bias Gemma rung and a pending label-permutation null (see Section 8).

CCS Concepts

• **Computing methodologies** $\rightarrow$ *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of “expert” subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the $\sim 0.5\text{--}0.6$ range toward $0.72\text{--}0.73$ for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and $n \leq 6$ point estimates. This round-II audit fixes the measurements, reports exact

permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

- (1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
- (2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at 0.05**. All four concentration metrics (normalized entropy H , Gini G , top-fractions t_5 and $t_{10\%}$) move monotonically with sparsity; dense models separate as a group. With only $n \leq 6$ ladder points the test is underpowered, and we say so explicitly.
- (3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation p -values, and an explicit model-set sensitivity audit reported for every subset so that “selecting the subsample that supports the claim” is impossible.
- (4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley flights) showing no demographic specialist structure.
- (5) Data/code release (`expt-bias-1/`).

Take-away. Take-away: none of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in our prior work [15]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9]) define the routers we instrument; recent surveys [10] catalogue the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [11], BBQ [12], holistic audits [13, 14]); our contribution measures not model-level bias magnitude but the *kernel location* of bias-relevant counterfactual shift within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries, we attach router hooks and represent every expert as a Shapley player [15]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (routing-contrast kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \quad (\text{normalized entropy}), \quad (1)$$

$$G = \frac{N + 1 - 2 \sum_{i=1}^N \text{cum}_i(p_{(i)})}{N} \quad (\text{Gini of } |\phi|), \quad (2)$$

$$t_5 = \sum_{i=1}^5 p_{(i)}, \quad (3)$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \quad (4)$$

where $p_{(i)}$ are the shares in decreasing order and $\text{cum}_i(\cdot)$ is the cumulative sum (so $0 \leq H \leq 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256\text{--}4608$ for MoE vs. $N = 32$ for dense). Lower H and higher G , t_5 , $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/ Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero ϕ (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$) and is used throughout.

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run’s `concentration_metrics.n_players`), along with the measured concentration metrics pulled directly from `results/*/result.json`.

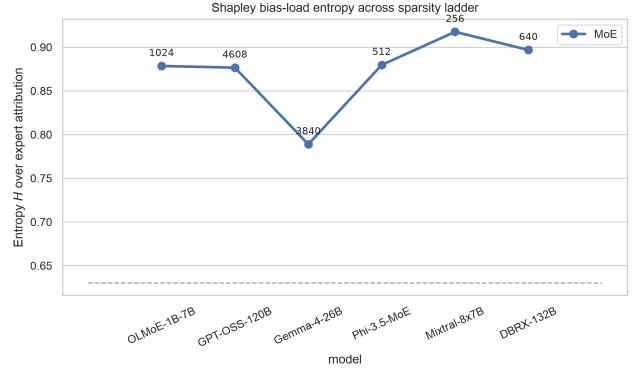


Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

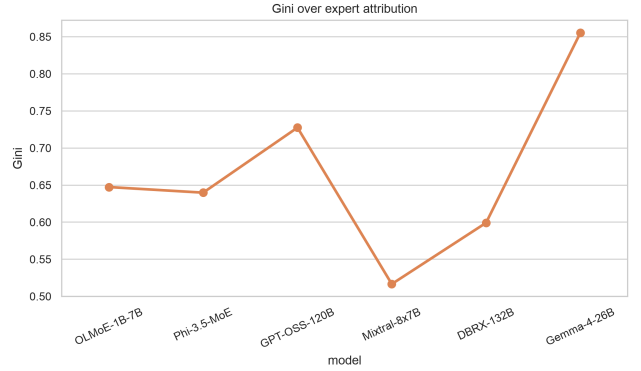


Figure 2: Gini of the expert attribution across the ladder.

5 Results

5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with k/N that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives $\rho_H = +0.754$ ($p = 0.106$, exact two-sided permutation); Gini gives $\rho_G = -0.754$ ($p = 0.106$). These are not significant at 0.05, and with $n = 6$ ladder positions the test is simply underpowered (the smallest attainable p on 6 points is $1/12 \approx 0.083$; on 5 points 0.10). Normalized entropy spans only 0.789–0.917 across the six rungs ($H \approx 0.88\text{--}0.92$ for the four un-flagged rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in k/N . By contrast, the smallest dense-to-MoE entropy gap is ≥ 0.064 ; the dense/MoE split is the robust contrast of the study.

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$	CI
OLMoE-1B-7B	16	1024	0.0156	0.878	0.647	0.092	0.510	[0.877, 0.886]
GPT-OSS-120B [†]	144	4608	0.0313	0.876	0.727	0.020	0.573	[0.875, 0.886]
Gemma-4-26B [‡]	240	3840	0.0625	0.789	0.855	0.051	0.767	[0.791, 0.802]
Phi-3.5-MoE	64	512	0.1250	0.880	0.640	0.095	0.469	[0.877, 0.892]
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.107	0.357	[0.913, 0.925]
DBRX-132B	160	640	0.2500	0.897	0.599	0.091	0.436	[0.897, 0.910]
Dense baselines (mean, $N=32$)	—	—	1.0	0.689	0.684	0.698	0.631	—

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03.h1.verdict.json` ladder rows). k/N = active-expert fraction (k from the verified audit table in `s03.h1.verdict.py`, N from stored `n_players`). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, `s04.bootstrap.cis.json`); intervals for Gini, t_5 , $t_{10\%}$ are reported in Table 2. [†]GPT-OSS-120B v1 is a valid capture ($n_{pairs} = 2000$); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ($\Delta H = -0.003$). [‡]Gemma’s mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	[0.877, 0.886]	[0.630, 0.649]	[0.084, 0.096]	[0.491, 0.514]
GPT-OSS-120B	[0.875, 0.886]	[0.707, 0.731]	[0.017, 0.021]	[0.540, 0.575]
Gemma-4-26B [‡]	[0.791, 0.802]	[0.841, 0.853]	[0.045, 0.051]	[0.736, 0.761]
Phi-3.5-MoE	[0.877, 0.892]	[0.610, 0.643]	[0.085, 0.100]	[0.442, 0.477]
Mixtral-8x7B	[0.913, 0.925]	[0.494, 0.529]	[0.098, 0.115]	[0.337, 0.368]
DBRX-132B	[0.897, 0.910]	[0.569, 0.603]	[0.072, 0.090]	[0.391, 0.434]

Table 2: 95% block-bootstrap confidence intervals for every MoE rung’s concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04.bootstrap.cis.json`). Intervals for H are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture ($n_{pairs} = 2000$). [‡]Gemma: see Table 1.

Model	H	Gini	t_5	$t_{10\%}$
OLMo-7B	0.719	0.693	0.687	0.605
Phi-3.5-Mini	0.758	0.613	0.625	0.528
Llama-2-7B	0.651	0.700	0.732	0.692
Llama-3.1-8B	0.630	0.730	0.749	0.698
Mean	0.689	0.684	0.698	0.631

Table 3: Dense-baseline measures (v0; `exp2-*/result.json`, one FFN layer player per transformer layer, $N = 32$). Bootstrap CIs are not derivable for dense rows: no per-pair ϕ payloads exist on disk (`s04.bootstrap.cis.json` covers the MoE ladder only), so these cells remain single-measure and are flagged CI:pending (Section 8).

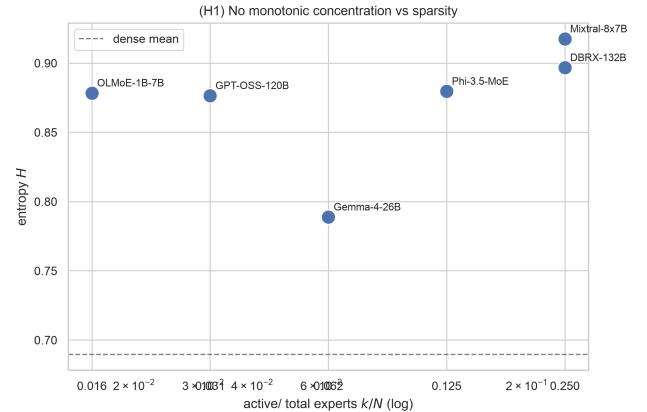


Figure 3: Entropy vs. active fraction k/N (log scale). H1 (sparser $\Rightarrow$ more concentrated) is directionally consistent ($\rho_H = +0.754$, exact two-sided permutation $p = 0.106$), but not statistically significant at 0.05 given $n = 6$ ladder points.

Take-away. Take-away: the sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

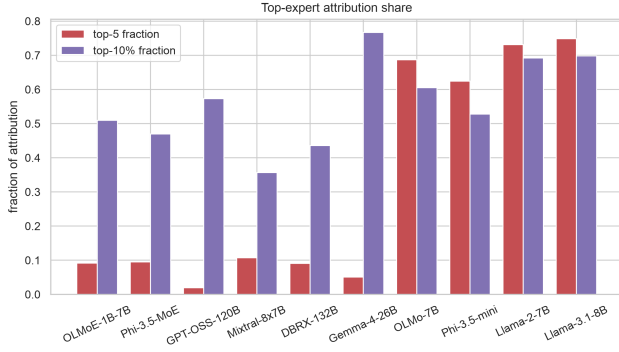


Figure 4: Top-5 and top-10% share across the ladder.

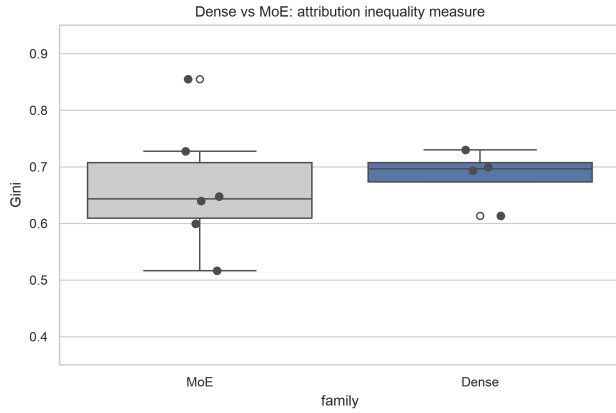


Figure 5: Dense vs. MoE Gini.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable “fixable” expert set.

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H : 0.630–0.758; mean 0.689, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it is a pointer-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs.

5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid 2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0–v1 rerun drift floors are $\text{floor}_H = 0.023$ and $\text{floor}_G = 0.049$ (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller than these floors (or than $2\times$ the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset’s verdict changes. Split-half shards (two sets of 2500 pairs) for Mixtral/DBRX agree to ≤ 0.004 in entropy.

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (paper-as-reported; per-prompt assignment not yet shipped, flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null’s p_{95} —cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation p -value for $n \leq 6$ (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = +0.754$, $p = 0.106$; $\rho_G = -0.754$, $p = 0.106$;
- 5 models (without GPT-OSS): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 5 models (paper ladder, without Gemma): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = +0.949$, $p = 0.167$; $\rho_G = -0.949$, $p = 0.167$;
- 3 distinct-sparsity rungs: $\rho_H = +1.000$, $p = 0.333$; $\rho_G = -1.000$, $p = 0.333$.

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at $p < 0.05$ in *any* subset: with $n \leq 6$ ladder points the attainable p -values are bounded below ($1/12 \approx 0.083$ at $n = 6$, 0.10 at $n = 5$, 0.167 at $n = 4$, 0.333 at $n = 3$), so the test is underpowered by construction. The directional signal is therefore consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; power is the limit, not the data.

6.4 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Twelve of thirteen models have per-pair ϕ payloads on disk and are reported in Table 2; only Gemma-4-27B (not part of the

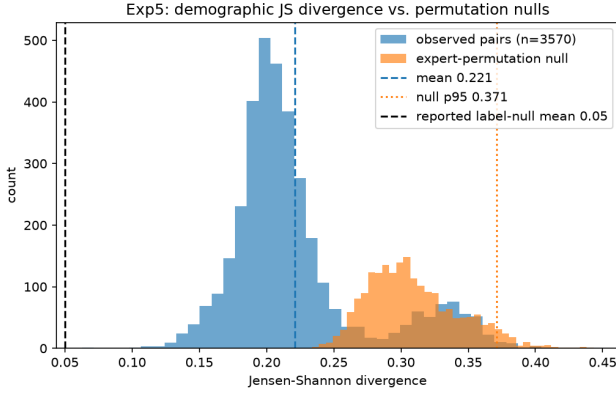


Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.

ladder) is MISSING. The cohort JS mean carries the same bootstrap treatment, 95% CI [0.206, 0.231].

6.5 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmo-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI [0.206, 0.231]), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null’s p_{95} , and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a “specialist”—consistent with the ladder finding. The most divergent pairs (e.g., Vietnam×software_developer, JS 0.27) do not form a coherent demographic structure.

Take-away. Take-away: bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after Gemma’s active-expert count was fixed (N_A : 960 → 240) and GPT-OSS-120B’s v1 capture was validated. Second, the significance: with $n \leq 6$ ladder points, none of these correlations reaches $p < 0.05$ (all-6 $p = 0.106$; the attainable p is bounded below by 1/12 on six points), so the claim “sparser routing concentrates bias attribution more” is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not

of the router. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both “measure” the same normalized concentration metrics (H , Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is ≈ 0) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy and Exp. 5 JS divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: “prune the bias experts” is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero ϕ directory that was discarded. The current v1 capture is a *valid* 2000-pair run ($H = 0.876$, Gini 0.727; CI H [0.875, 0.886]) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$, $\Delta G = +0.004$).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; twelve of thirteen model payloads are on disk.
- **Gemma-4-26B is a null-bias model — flag: null-bias.** Its mean bias gap ≈ 0 and 47% of its ϕ are exactly zero; its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.
- **Exact permutation p on $n \leq 6$ — flag: small- n .** Spearman p -values are exact two-sided permutation probabilities, which are bounded below by construction (1/12 ≈ 0.083 on 6 points, 0.10 on 5, 0.167 on 4, 0.333 on 3). No subset reaches $p < 0.05$; we therefore report

drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.

- **Label-permutation null** — **flag:** `null:label-pending`. The reported-label null in `s02` was computed as-reported; the per-prompt assignment is not yet shipped. The expert-index null used here is on-disk and reproducible.
- **Top- k plumbing** — **flag:** `hooks:pending`. k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX's N_A and any future config divergence may alter k/N assignment.

9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6 $\rho_H = +0.754$ ($p = 0.106$), paper-valid-4 $\rho_H = +0.949$ ($p = 0.167$), unique-sparsity-3 $\rho_H = +1.000$ ($p = 0.333$)—but that does not reach statistical significance at 0.05 on $n \leq 6$ ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

Take-away. Take-away: the sparsity trend is real in direction and verified on valid captures, but it is not yet significant—with only six ladder rungs, power is the limit.

Data & Code

- Runs and per-pair Shapley payloads: `expt-bias-1/results/*/result.json`.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py` (all under `stats_analysis/scripts/`).
- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.

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